Lecture 9: Improper Integrals

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Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Know what is meant for an improper integral with unbounded domain to converge or diverge.
- ☐ Know what the values of p such that the p-integrals converge or diverge.
- ☐ Know what is meant for an improper integral involving integrand with vertical asymptote to converge or diverge.
- ☐ Use the direct comparison test to tell whether an improper integral converges or diverges when possible and know the limitations of the theorem.
- ☐ Use the limit comparison test to tell whether an improper integral converges or diverges when possible and know the limitations of the theorem.
- ☐ Identify the p-integral to be used for the comparison test even if the question does not suggest it.

Introduction

So far, we have considered definite integrals of the form $I = \int_a^b f(x)\,dx$, where the integrand f is continuous over the interval $[a, b]$.

Such integral I is necessarily a finite number and is called proper integrals.

Let $I = \int_a^b f(x)\,dx$ be a definite integral,

I is proper when

I is improper when

Improper Integral as the Area Under the Curve

An improper integral of a non-negative function can be related to the area under the curve.

If the area under the curve is finite, then we say the improper integral converges.

Otherwise, we say the improper integral diverges.

Type I Improper Integrals

Integrals with infinite limit of integrations are improper integrals of Type I:

- If $f(x)$ is continuous on $[a, \infty)$, then

$$\int_a^{\infty} f(x)\,dx = \lim_{b\to\infty} \int_a^b f(x)\,dx.$$

- If $f(x)$ is continuous on $(-\infty, b]$, then

$$\int_{-\infty}^b f(x)\,dx = \lim_{a\to-\infty} \int_a^b f(x)\,dx.$$

- If $f(x)$ is continuous on $(-\infty, \infty)$, then

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^c f(x)\,dx + \int_c^{\infty} f(x)\,dx,$$

  where c is any real number.

In each case, if the limit exists and is finite, we say that the improper integral converges and that the limit is the value of the improper integral. If the limit fails to exist, the improper integral diverges.

Example

Is the area under the curve $y = \frac{\ln x}{x^2}$ from $x = 1$ to $x = \infty$ finite? If so, what is its value?

Example

Evaluate $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$

Example

For what values of p does the integral $\int_1^\infty \frac{dx}{x^p}$ converge? When the integral converge, what is its value?

Type II Improper Integrals

Integrals of functions that become infinite at a point within the interval of integration are improper integrals of Type II:

- If $f(x)$ is continuous on $(a, b]$ and discontinuous at a, then
$$\int_a^b f(x)\,dx = \lim_{c \to a^+} \int_c^b f(x)\,dx.$$
- If $f(x)$ is continuous on $[a, b)$ and discontinuous at b, then
$$\int_a^b f(x)\,dx = \lim_{c \to b^-} \int_a^c f(x)\,dx.$$
- If $f(x)$ discontituous at c where $a < c < b$, and is continuous on $[a, c) \cup (c, b]$, then
$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx,$$

In each case, if the limit exists and is finite, we say that the improper integral converges and that the limit is the value of the improper integral. If the limit fails to exist, the improper integral diverges.

Example

Investigate the convergence of $\int_0^1 \frac{1}{1-x}\,dx$.

Example

Evaluate $\int_0^3 \frac{dx}{(x-1)^{2/3}}$.

Example

Evaluate $\int_2^{\infty} \frac{x+3}{(x-1)(x^2+1)}\,dx$.

Example

Evaluate $\int_{-1}^{2} \frac{1}{x^2}\, dx$

Tests for Convergence and Divergence

Sometimes, we may not be able to evaluate the improper integral directly.

However, we may still be able to determine whether the integral converges or diverges by comparing it one integral whose convergence properties are known.

Two comparison test:

- Direct Comparison Test
- Limit Comparison Test

These tests only tell us if the integral converges or diverges. To find (or estimate) its value, if converges, we need to use other (numerical) methods to approximate the value.

In many cases, it is helpful to compare your given integral with that of a p-integral.

Direct Comparison Test

Let f and g be continuous on $[a, \infty)$ with $0 \leq f(x) \leq g(x)$ for all $x \leq a$. Then

1. if $\int_a^\infty g(x)\,dx$ converges, then $\int_a^\infty f(x)\,dx$ also converges.
2. if $\int_a^\infty f(x)\,dx$ diverges, then $\int_a^\infty g(x)\,dx$ also diverges.

Direct Comparison Test

$\int_1^{\infty} e^{-x^2}\,dx$ converges

Direct Comparison Test

$$\int_1^{\infty} \frac{\sin^2 x}{x^2}\, dx \text{ converges}$$

Direct Comparison Test

$$\int_1^{\infty} \frac{1}{\sqrt{x^2 - 0.1}}\, dx \text{ diverges}$$

Direct Comparison Test

$\int_0^{\pi/2} \frac{\cos x}{\sqrt{x}}\, dx$ converges

Direct Comparison Test

Do not use the Direct Comparison Test to test $\int_a^\infty f(x)\,dx$ when

(a) $f(x) \leq g(x)$ and $\int_a^\infty g(x)\,dx$ diverges, or

(b) $g(x) \leq f(x)$ and $\int_a^\infty g(x)\,dx$ converges.

For example, to determine if $\int_1^\infty \frac{1-e^{-x}}{x}\,dx$ we cannot use Direct Comparison Test to compare it with $\int_1^\infty \frac{1}{x}$ even though we know $\frac{1-e^{-x}}{x} \leq \frac{1}{x}$. This is because the integral $\int_1^\infty \frac{1}{x}$ diverges, and we cannot say anything about $\int_1^\infty \frac{1-e^{-x}}{x}\,dx$. Instead, we can use the Limit Comparison Test.

Limit Comparison Test

If the positive functions f and g are continuous on $[a, \infty)$, and if

$$\lim_{x \to \infty} \frac{f(x)}{g(x)} = L, \ 0 < L < \infty,$$

then

$$\int_a^\infty f(x)\,dx \ \text{ and } \ \int_a^\infty g(x)\,dx$$

either both converge or both diverge.

Limit Comparison Test

Show that

$$\int_1^\infty \frac{dx}{1+x^2}$$

converges by comparison with $\int_1^\infty (1/x^2)\,dx$. Find and compare the values of two integrals.

Limit Comparison Test

Investigate the convergence of $\int_1^{\infty} \frac{1 - e^{-x}}{x}\, dx$.

Limit Comparison Test

Investigate the convergence of $\displaystyle\int_2^{\infty} \frac{\sqrt{3x^2+\sin x}}{x^4-2}\,dx$.